

A Fixed-Parameter Criterion for Least (a, b) -Majorized Elements

Verification note

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Abstract

Fix one positive scaling vector $a \in \mathbb{R}_{>0}^n$ and one shift vector $b \in \mathbb{R}^n$. This note proves an exact finite criterion for a nonempty feasible set $C \subseteq \mathbb{R}^n$ to contain a least fixed- (a, b) -majorized element. Equivalently, it characterizes when some $x^* \in C$ simultaneously minimizes, over C , every objective

$$x \mapsto \sum_{i=1}^n a_i \phi\left(\frac{x_i + b_i}{a_i}\right),$$

as $\phi : \mathbb{R} \rightarrow \mathbb{R}$ ranges over continuous convex functions. The criterion is that C lie in a constant-sum hyperplane and that finitely many weighted upper-rank functions have a common minimizer on C .

The result is a complete solution of this fixed-parameter existence problem. It is not a solution of the stronger structural problem in Schoot Uiterkamp's paper, where compact convex sets having least (a, b) -majorized elements for all choices of a and b are characterized as base polyhedra. Relative to that all-parameters structural problem, the result here is partial progress: it is a finite certificate for a prescribed pair (a, b) , not a base-polyhedral characterization.

1 Status relative to the source paper

Let $N := \{1, \dots, n\}$. For $S \subseteq N$ write

$$a(S) := \sum_{i \in S} a_i, \quad x(S) := \sum_{i \in S} x_i, \quad A := a(N).$$

Schoot Uiterkamp [3] studies least majorized elements and the generalized notion of least (a, b) -majorized elements. One of the main results there is structural: within compact convex sets, having least (a, b) -majorized elements for every positive vector a and every shift vector b characterizes base polyhedra.

The result proved below is different. The pair (a, b) is fixed once and for all. For that fixed pair, we give a necessary and sufficient finite criterion for the existence of a least element in an arbitrary nonempty set $C \subseteq \mathbb{R}^n$. No convexity, compactness, closedness, or polyhedrality is assumed. Thus the theorem proves the fixed-parameter problem exactly, but it does not imply that C is a base polyhedron and it does not settle the all-parameters base-polyhedron characterization. Indeed, for one fixed pair (a, b) , very irregular sets can have a least element; see Proposition 6.1 and Remark 6.2.

2 Definitions

Throughout, $a \in \mathbb{R}_{>0}^n$ and $b \in \mathbb{R}^n$ are fixed. Define

$$\Psi_{a,b}(x)_i := \frac{x_i + b_i}{a_i}, \quad i \in N.$$

For $z, w \in \mathbb{R}^n$, write $z \preceq_a w$ if

$$\sum_{i=1}^n a_i z_i = \sum_{i=1}^n a_i w_i \quad (1)$$

and

$$\sum_{i=1}^n a_i \phi(z_i) \leq \sum_{i=1}^n a_i \phi(w_i) \quad \text{for every continuous convex } \phi : \mathbb{R} \rightarrow \mathbb{R}. \quad (2)$$

Equivalently, the weighted atomic measure $\sum_i a_i \delta_{z_i}$ is below $\sum_i a_i \delta_{w_i}$ in convex order.

For $x, y \in \mathbb{R}^n$, write

$$x \preceq_{a,b} y \quad \text{if and only if} \quad \Psi_{a,b}(x) \preceq_a \Psi_{a,b}(y).$$

Since

$$\sum_i a_i \Psi_{a,b}(x)_i = x(N) + b(N),$$

the equality condition in $x \preceq_{a,b} y$ is simply $x(N) = y(N)$.

Definition 2.1. Let $C \subseteq \mathbb{R}^n$ be nonempty. A point $x^* \in C$ is a *least fixed- (a, b) -majorized element* of C if

$$x^* \preceq_{a,b} y \quad \text{for every } y \in C.$$

The set of all such points is denoted by $\mathcal{L}_{a,b}(C)$.

Equivalently, $x^* \in C$ is least fixed- (a, b) -majorized if and only if

$$\sum_{i=1}^n a_i \phi(\Psi_{a,b}(x^*)_i) \leq \sum_{i=1}^n a_i \phi(\Psi_{a,b}(y)_i) \quad (y \in C)$$

for every continuous convex ϕ . The equality of total sums follows automatically from applying this inequality to the affine convex functions $\phi(t) = t$ and $\phi(t) = -t$.

3 Weighted upper-rank functions

For $r \in [0, A]$ and $z \in \mathbb{R}^n$, define

$$U_r(z) := \max \left\{ \sum_{i=1}^n u_i z_i : 0 \leq u_i \leq a_i, \sum_{i=1}^n u_i = r \right\}. \quad (3)$$

Thus $U_r(z)$ is the sum of the largest r units of mass in the weighted multiset that places mass a_i at z_i . If $a_i = 1$ for all i and $r = k \in \{0, 1, \dots, n\}$, then $U_k(z)$ is the sum of the k largest coordinates of z .

Set

$$\mathcal{R}(a) := \{a(S) : S \subseteq N\}, \quad \mathcal{R}^\circ(a) := \mathcal{R}(a) \setminus \{0, A\}.$$

Lemma 3.1 (weighted Hardy–Littlewood–Polya criterion). *For $z, w \in \mathbb{R}^n$, the following are equivalent.*

(i) $z \preceq_a w$.

(ii) $\sum_i a_i z_i = \sum_i a_i w_i$ and

$$U_r(z) \leq U_r(w) \quad \text{for every } r \in [0, A].$$

(iii) $\sum_i a_i z_i = \sum_i a_i w_i$ and

$$U_r(z) \leq U_r(w) \quad \text{for every } r \in \mathcal{R}(a).$$

Proof. For $z \in \mathbb{R}^n$, define the call function

$$K_z(t) := \sum_{i=1}^n a_i (z_i - t)_+, \quad t \in \mathbb{R}.$$

For every $t \in \mathbb{R}$,

$$K_z(t) = \max_{0 \leq u_i \leq a_i} \sum_{i=1}^n u_i (z_i - t) = \max_{0 \leq r \leq A} (U_r(z) - rt). \quad (4)$$

Conversely, for every $r \in [0, A]$,

$$U_r(z) = \inf_{t \in \mathbb{R}} (K_z(t) + rt). \quad (5)$$

Indeed, for every feasible u in (3) and every t ,

$$\sum_i u_i z_i = \sum_i u_i (z_i - t) + rt \leq \sum_i a_i (z_i - t)_+ + rt = K_z(t) + rt,$$

so $U_r(z) \leq \inf_t (K_z(t) + rt)$. Equality is obtained by taking t to be a weighted r -quantile of the multiset $\{z_i\}$ with masses a_i : choose $u_i = a_i$ when $z_i > t$, choose $u_i = 0$ when $z_i < t$, and split the remaining mass among indices with $z_i = t$ so that $\sum_i u_i = r$.

Assume first that the weighted sums of z and w are equal and that $U_r(z) \leq U_r(w)$ for every $r \in [0, A]$. By (4),

$$K_z(t) \leq K_w(t) \quad \text{for every } t \in \mathbb{R}.$$

Every piecewise-linear convex function on an interval containing all coordinates of z and w is an affine function plus a nonnegative linear combination of hinge functions $(s - t)_+$. Since affine parts cancel by equality of the weighted sums and the hinge inequalities follow from $K_z \leq K_w$, inequality (2) holds for all piecewise-linear convex functions on that interval. Uniform approximation of continuous convex functions on the interval by piecewise-linear convex functions gives (2) for every continuous convex ϕ . Hence $z \preceq_a w$.

Conversely, if $z \preceq_a w$, then applying (2) to the convex function $s \mapsto (s - t)_+$ gives $K_z(t) \leq K_w(t)$ for every t . Using (5),

$$U_r(z) = \inf_t (K_z(t) + rt) \leq \inf_t (K_w(t) + rt) = U_r(w)$$

for every $r \in [0, A]$. This proves the equivalence of (i) and (ii).

It remains to replace the continuum of ranks by the finite set $\mathcal{R}(a)$. Fix z and order the indices so that

$$z_{\pi(1)} \geq z_{\pi(2)} \geq \cdots \geq z_{\pi(n)}.$$

Put $A_0 := 0$ and $A_k := \sum_{h=1}^k a_{\pi(h)}$. As a function of r , $U_r(z)$ is linear on each interval $[A_{k-1}, A_k]$, and every breakpoint A_k belongs to $\mathcal{R}(a)$. The same is true for w . If

$$\mathcal{R}(a) = \{\gamma_0 < \gamma_1 < \cdots < \gamma_m\},$$

then both $r \mapsto U_r(z)$ and $r \mapsto U_r(w)$ are linear on every interval $[\gamma_{j-1}, \gamma_j]$. Therefore $U_r(z) \leq U_r(w)$ at all $r \in \mathcal{R}(a)$ if and only if it holds for all $r \in [0, A]$. $\square$

4 The fixed-parameter characterization

For a nonempty set $C \subseteq \mathbb{R}^n$ and $r \in [0, A]$, define

$$m_r(C) := \inf_{x \in C} U_r(\Psi_{a,b}(x)). \quad (6)$$

Theorem 4.1 (finite characterization of least fixed- (a, b) elements). *Let $C \subseteq \mathbb{R}^n$ be nonempty. Then C admits a least fixed- (a, b) -majorized element if and only if the following two conditions hold.*

- (1) *C is contained in a constant-sum hyperplane: there exists $\tau \in \mathbb{R}$ such that*

$$x(N) = \tau \quad \text{for every } x \in C.$$

- (2) *The finitely many upper-rank objectives have a common minimizer:*

$$C \cap \bigcap_{r \in \mathcal{R}^\circ(a)} \arg \min_{x \in C} U_r(\Psi_{a,b}(x)) \neq \emptyset.$$

When these conditions hold, the set of all least fixed- (a, b) -majorized elements is exactly

$$\mathcal{L}_{a,b}(C) = C \cap \bigcap_{r \in \mathcal{R}^\circ(a)} \arg \min_{x \in C} U_r(\Psi_{a,b}(x)). \quad (7)$$

Equivalently, $x^* \in C$ is least fixed- (a, b) -majorized if and only if $x(N)$ is constant on C and

$$U_r(\Psi_{a,b}(x^*)) = m_r(C) \quad \text{for every } r \in \mathcal{R}^\circ(a). \quad (8)$$

Proof. Suppose first that $x^* \in C$ is least fixed- (a, b) -majorized. Then for every $y \in C$,

$$\Psi_{a,b}(x^*) \preceq_a \Psi_{a,b}(y).$$

Applying the weighted-sum equality in the definition of $\preceq_a$ gives

$$x^*(N) + b(N) = y(N) + b(N),$$

and therefore $y(N) = x^*(N)$ for every $y \in C$. Thus C lies in a constant-sum hyperplane.

Now put $z^* := \Psi_{a,b}(x^*)$ and $w := \Psi_{a,b}(y)$. By Lemma 3.1,

$$U_r(z^*) \leq U_r(w) \quad \text{for every } r \in \mathcal{R}(a)$$

for every $y \in C$. Since $x^* \in C$, this says exactly that x^* minimizes $U_r(\Psi_{a,b}(x))$ over C for every $r \in \mathcal{R}^\circ(a)$.

Conversely, assume that $C \subseteq \{x : x(N) = \tau\}$ and that $x^* \in C$ minimizes all the objectives $U_r(\Psi_{a,b}(x))$ with $r \in \mathcal{R}^\circ(a)$. Let $y \in C$ be arbitrary, and again set $z^* := \Psi_{a,b}(x^*)$ and $w := \Psi_{a,b}(y)$. The constant-sum condition gives

$$\sum_i a_i z_i^* = x^*(N) + b(N) = y(N) + b(N) = \sum_i a_i w_i.$$

The common-minimizer condition gives

$$U_r(z^*) \leq U_r(w) \quad \text{for every } r \in \mathcal{R}^\circ(a).$$

At $r = 0$ the two sides are both zero, and at $r = A$ the two sides are both equal to the common weighted sum. Hence all inequalities required in Lemma 3.1(iii) hold, so $z^* \preceq_a w$. Therefore $x^* \preceq_{a,b} y$. Since $y \in C$ was arbitrary, x^* is least fixed- (a, b) -majorized. Formula (7) follows from the two implications just proved. $\square$

5 A finite linear certificate

The functions appearing in Theorem 4.1 are maxima of finitely many affine functions. For $0 < r < A$, define

$$\mathcal{E}_r := \{(S, j) : S \subseteq N \setminus \{j\}, a(S) \leq r \leq a(S) + a_j\}.$$

The extreme points of the polytope

$$\{u \in \mathbb{R}^n : 0 \leq u_i \leq a_i, u(N) = r\}$$

have at most one coordinate strictly between its bounds. Hence every such extreme point has the form

$$u_i = a_i \ (i \in S), \quad u_j = r - a(S), \quad u_i = 0 \ (i \notin S \cup \{j\})$$

for some $(S, j) \in \mathcal{E}_r$, and therefore

$$U_r(z) = \max_{(S, j) \in \mathcal{E}_r} \left(\sum_{i \in S} a_i z_i + (r - a(S)) z_j \right). \quad (9)$$

In the original variables this becomes

$$U_r(\Psi_{a,b}(x)) = \max_{(S, j) \in \mathcal{E}_r} \left(\sum_{i \in S} (x_i + b_i) + \frac{r - a(S)}{a_j} (x_j + b_j) \right). \quad (10)$$

Thus Theorem 4.1 can be written entirely as a finite collection of scalar minimization and feasibility conditions involving affine inequalities.

Corollary 5.1 (polyhedral feasibility test). *Suppose C is a nonempty polyhedron described by finitely many linear inequalities. Then existence of a least fixed- (a, b) -majorized element is decidable by finitely many linear programs.*

More explicitly, first check whether $x(N)$ is constant on C . If it is not, then no least fixed- (a, b) element exists. If it is, solve, for each $r \in \mathcal{R}^\circ(a)$, the epigraph linear program

$$\begin{aligned} & \text{minimize} && t \\ & \text{subject to} && x \in C, \\ & && \sum_{i \in S} (x_i + b_i) + \frac{r - a(S)}{a_j} (x_j + b_j) \leq t, \quad (S, j) \in \mathcal{E}_r. \end{aligned}$$

If any one of these programs is unbounded below, then no least fixed- (a, b) element exists. Otherwise let m_r denote its finite optimal value. Then C has a least fixed- (a, b) -majorized element if and only if the finite linear system

$$x \in C, \quad \sum_{i \in S} (x_i + b_i) + \frac{r - a(S)}{a_j} (x_j + b_j) \leq m_r$$

for all $r \in \mathcal{R}^\circ(a)$ and all $(S, j) \in \mathcal{E}_r$ is feasible. Every feasible solution is a least fixed- (a, b) -majorized element.

Proof. The epigraph formulation is exactly (10). A finite optimal value of a linear program over a nonempty polyhedron is attained. By Theorem 4.1, after the constant-sum condition has been checked, the only remaining condition is the existence of a point attaining all the finitely many values m_r simultaneously. The displayed system is precisely that common-argmin condition. If some $m_r = -\infty$, no point of C can attain it, so no common minimizer exists. $\square$

Remark 5.2. The test is finite but not claimed to be polynomial-time. The set $\mathcal{R}(a)$ can have 2^n distinct ranks, and the families $\mathcal{E}_r$ may also be exponentially large.

6 Consequences and special cases

Proposition 6.1 (the whole hyperplane has a fixed least element). *Fix $a \in \mathbb{R}_{>0}^n$ and $b \in \mathbb{R}^n$. Let*

$$H_R := \{x \in \mathbb{R}^n : x(N) = R\}.$$

Define

$$\bar{x}_i := a_i \frac{R + b(N)}{a(N)} - b_i, \quad i \in N. \quad (11)$$

Then $\bar{x} \in H_R$ and $\bar{x}$ is a least fixed- (a, b) -majorized element of H_R . Consequently every subset $C \subseteq H_R$ with $\bar{x} \in C$ has $\bar{x}$ as a least fixed- (a, b) -majorized element.

Proof. The definition gives $\bar{x}(N) = a(N)(R + b(N))/a(N) - b(N) = R$. Moreover,

$$\Psi_{a,b}(\bar{x})_i = \frac{R + b(N)}{a(N)}$$

is independent of i . For any $y \in H_R$ and any continuous convex ϕ , Jensen's inequality with weights $a_i/a(N)$ gives

$$\begin{aligned} \sum_i a_i \phi(\Psi_{a,b}(y)_i) &\geq a(N) \phi\left(\frac{1}{a(N)} \sum_i a_i \Psi_{a,b}(y)_i\right) \\ &= a(N) \phi\left(\frac{R + b(N)}{a(N)}\right) = \sum_i a_i \phi(\Psi_{a,b}(\bar{x})_i). \end{aligned}$$

Thus $\bar{x} \preceq_{a,b} y$ for every $y \in H_R$, and the subset statement is immediate. $\square$

Remark 6.2 (why the fixed-parameter result is not structural). For one fixed pair (a, b) , least elements can occur in very non-polyhedral sets. For instance, if $n \geq 3$ and $V := \{v \in \mathbb{R}^n : v(N) = 0\}$, then

$$C := \{\bar{x} + v : v \in V, \|v\|_2 \leq 1\}$$

is compact and convex but is not a polyhedron, while Proposition 6.1 shows that it has a least fixed- (a, b) -majorized element. This illustrates why Theorem 4.1 should be viewed as a fixed-parameter certificate, not as a replacement for the all-parameter base-polyhedron characterization in [3].

Taking $b = 0$ in Theorem 4.1 gives the scaled-only characterization.

Corollary 6.3 (fixed scaling, no shift). *Fix $a \in \mathbb{R}_{>0}^n$ and set $b = 0$. A nonempty feasible set $C \subseteq \mathbb{R}^n$ admits an element x^* satisfying*

$$\sum_{i=1}^n a_i \phi\left(\frac{x_i^*}{a_i}\right) \leq \sum_{i=1}^n a_i \phi\left(\frac{y_i}{a_i}\right) \quad (y \in C, \phi \text{ continuous convex})$$

if and only if C lies in a constant-sum hyperplane and the functions

$$x \longmapsto \max \left\{ \sum_{i=1}^n u_i \frac{x_i}{a_i} : 0 \leq u_i \leq a_i, \sum_i u_i = r \right\}, \quad r \in \mathcal{R}^\circ(a),$$

have a common minimizer on C . The set of all such x^ is exactly the common argmin set.*

When $a_i = 1$ for all i and $b = 0$, $\mathcal{R}(a) = \{0, 1, \dots, n\}$ and

$$U_k(x) = x_{[1]} + \dots + x_{[k]},$$

where $x_{[1]} \geq \dots \geq x_{[n]}$ are the coordinates of x sorted in nonincreasing order. Thus the theorem reduces to the familiar ordinary-majorization test.

Corollary 6.4 (ordinary majorization). *A nonempty set $C \subseteq \mathbb{R}^n$ contains a least majorized element if and only if C is contained in a hyperplane $x(N) = \tau$ and there is $x^* \in C$ that simultaneously minimizes*

$$x_{[1]} + \dots + x_{[k]}, \quad k = 1, \dots, n-1,$$

over C . Such common minimizers are precisely the least majorized elements of C .

Remark 6.5. No convexity, compactness, closedness, or polyhedrality of C is used in Theorem 4.1. Those assumptions only affect whether the finitely many scalar infima in (6) are attained and how the common-argmin condition can be checked.

References

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